CST545 - Embedded Development

CST 545 (Embedded Development) Syllabus

Course Information

Course Details

- **Course Title:** Embedded Development
- **Course Number:** CST 545
- **Meeting Location:** Physical
- **Units:** 2

Course Description

This course introduces embedded systems development, focusing on programming directly on hardware and handling low-level I/O interfacing. Key topics include memory-mapped I/O, interfacing with General Purpose Input/Output (GPIO), and working with development boards. The course also covers considerations for power consumption in embedded systems. Through hands-on projects, students will gain practical experience designing and implementing programs that interact directly with hardware components.

Materials

- Course textbook: "Programming Embedded Systems: With C and GNU Development Tools" by Michael Barr and Anthony Massa

Technology Requirements

Students must have access to: - A computer (not "chromebook" or similar) with reliable internet connection - Access to Canvas learning management system - Embedded development boards (Arduino, Raspberry Pi, or similar microcontroller platforms) - Basic electronic components and breadboards for circuit prototyping - Access to oscilloscope and multimeter for hardware debugging

Course Objectives

At the end of this class, you should be able to:

1. Program embedded systems with direct hardware interfacing and low-level I/O operations
2. Implement memory-mapped I/O and GPIO interfacing techniques
3. Design power-efficient embedded applications with appropriate consumption considerations
4. Develop and debug embedded programs using development boards and hardware tools

Grading

There are three groups of assignments, briefly described in the table below. **Getting below a 40% in any of these three groups will result in failing the class.**

Assignment Kind	Value
Projects	45%
Exams	25%
Participation	30%

Projects

Projects will take the form of hardware prototyping assignments, embedded system implementations, and circuit design challenges. Students will program microcontrollers, interface with sensors and actuators, and develop power-efficient embedded applications.

Exams

Exams will cover material from class and from any assigned reading, as well as draw from knowledge gained during projects.

Participation

Class is expected to be highly interactive, with students actively answering and asking questions, as well as participating in discussion groups. Therefore, attendance is required, as well as participation in all in-class activities. These activities may take the form of in-class discussions, brainstorming sessions, stand-ups, as well as making active progress on projects during class time.

Grade Assignment

Grades as assigned based on the standard ranges, which are outlined below.

Grade Range	Letter Grade
$[97, \infty)$	A+
$[93, 97)$	A
$[90, 93)$	A-
$[87, 90)$	B+
$[83, 87)$	B
$[80, 83)$	B-
$[77, 80)$	C+
$[73, 77)$	C
$[70, 73)$	C-
$[60, 70)$	D
$[0, 60)$	F

Syllabus Change Policy

This syllabus is subject to change at the instructor's discretion to enhance learning or accommodate unforeseen circumstances. Students will be notified of any substantive changes (such as changes to due dates, point values of assignments, or major policy modifications) via Canvas announcement and/or email at least one week in advance when possible.

The process for syllabus changes: 1. Instructor identifies need for change 2. Revised syllabus section is prepared 3. Students are notified via Canvas and email 4. Change takes effect after notification period

Students are responsible for staying current with any announced changes to the syllabus.